

HARLEEN KOUR

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Professional Summary

Recent B.Tech graduate in Information Technology (CGPA: 8.0, result awaited) with hands-on experience in Machine Learning, Artificial Intelligence, and Data Analytics. Developed **BGSBU Smart Assistant Chatbot** as a major project — an ML-powered university query assistant with hybrid retrieval (BM25 + Sentence Transformers), 5,447 QA pairs, and 84% user satisfaction. Completed internship at IIT Jammu and data analytics simulation at Deloitte. Proficient in Python, scikit-learn, NLP, and SQL, seeking fresher-level roles in AI, Machine Learning, Data Analytics, or Web Development.

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, SQL, PHP, HTML, CSS
- **ML/AI:** scikit-learn, NLTK, NLP, Sentence Transformers, BM25, Vector Embeddings, Random Forest, SVM, Linear/Logistic Regression, K-Means, Supervised/Unsupervised Learning, Model Deployment, Data Pipeline
- **Deep Learning:** Hugging Face Transformers, all-MiniLM-L6-v2, TensorFlow, PyTorch
- **Web/Full Stack:** React 18, TypeScript, Node.js, FastAPI, REST APIs, MySQL, PostgreSQL, JWT Authentication
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn, Power BI, EDA, Feature Engineering, Data Cleaning
- **Dev Tools:** Git, GitHub, Jupyter Notebook, VS Code, Ubuntu, Docker
- **Soft Skills:** Analytical Thinking, Problem Solving, Team Collaboration, Technical Presentation, Continuous Learning

Internship & Work Experience

- **Data Science & Artificial Intelligence Intern** **Jun 2025 – Aug 2025**
Indian Institute of Technology (IIT Jammu) – Summer School 2025 *On-site*
 - Built and benchmarked 3 ML models (Linear Regression, Random Forest, SVM) achieving up to 91% F1-score, AUC-ROC, and precision/recall.
 - Developed end-to-end ML pipelines using Python, Pandas, NumPy, and scikit-learn with automated feature engineering and cleaning.
 - Applied EDA on 3 datasets (10,000+ records each), identifying predictive features that improved model performance by 12–18%.
- **Data Analytics Virtual Job Simulation** **Jun 2025 – Aug 2025**
Deloitte (via Forage) *Remote*
 - Analysed structured business datasets to detect financial anomalies using Python and Excel-based pipelines, delivering findings in a structured consulting report.
 - Applied analytical thinking to 3 business case scenarios, producing data-backed recommendations aligned with consulting methodology.
- **Machine Learning Training Program** **2024**
Winnovation *On-site*
 - Implemented 5+ ML algorithms including KNN, SVM, K-Means, Linear/Logistic Regression, and Decision Trees with hands-on performance tuning.

Projects

- **BGSBU Smart Assistant Chatbot** [Major Project] **2025 – 2026**
Python, FastAPI, React 18, TypeScript, NLP, Sentence Transformers, BM25, scikit-learn
 - Designed and deployed an ML-powered university query assistant serving 2,000+ students, handling admissions, fees, exams, and hostel queries with 84% user satisfaction.
 - Engineered a hybrid retrieval engine combining BM25 and Sentence Transformers (all-MiniLM-L6-v2, 384-dim) with Reciprocal Rank Fusion, improving precision by ~22%.
 - Built a 5,447 Q&A corpus across 284 intents using automated PDF ingestion and a 5-pass cleaning pipeline, achieving 91% intent classification accuracy.
 - Implemented continuous learning where feedback is injected into training data and models retrained, improving satisfaction from 71% to 84% over 3 iterations.
 - Developed a React 18 + TypeScript frontend with real-time confidence visualisation, quick-question chips, and live feedback integrated via FastAPI REST APIs.

- Sentiment Analysis System** 2024
 - Python, NLTK, scikit-learn, TF-IDF, SVM, Pandas*
 - Built an NLP pipeline to classify 50,000+ customer reviews into positive, negative, and neutral sentiment using TF-IDF and SVM, achieving 88% accuracy.
 - Applied tokenisation, stopwords removal, and lemmatisation, reducing feature dimensionality by 35% while maintaining performance.
- Customer Churn Prediction** 2024
 - Python, scikit-learn, Random Forest, SMOTE, Pandas, Seaborn*
 - Developed a Random Forest classifier for a 21,000-record telecom dataset, achieving AUC-ROC of 0.87 and F1-score of 0.83 on the minority class.
 - Resolved 78:22 class imbalance using SMOTE, improving minority-class recall from 61% to 83% and reducing false negatives by ~36%.
- QR Code-Based Complaint Management System** 2024
 - Node.js, MySQL, REST API, JavaScript, Full Stack*
 - Built a full-stack QR code complaint reporting system for campus management, reducing submission time from ~24 hours to real-time mobile reporting.
 - Designed 8 RESTful API endpoints with JWT-style validation, supporting concurrent submissions from 500+ users.
- Stock Market Trend Prediction** 2024
 - Python, scikit-learn, Linear Regression, Random Forest, Matplotlib*
 - Compared Linear Regression and Random Forest on 5 years of stock data; Random Forest reduced RMSE by 31% versus the linear baseline.
- WhatsApp Chat Analyser** 2024
 - Python, Pandas, Regex, Matplotlib, Seaborn*
 - Parsed and analysed 120,000+ messages from exported WhatsApp data to extract heatmaps, word frequencies, emoji usage, and response-time trends.

Education

- Bachelor of Technology – Information Technology & Engineering** 2022 – May 2026 (Expected)
 - Baba Ghulam Shah Badshah University (BGSBU), Rajouri, J&K* CGPA: 8.0 / 10
 - Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering
 - Electives: Machine Learning, Data Science, Artificial Intelligence, Web Technologies
 - Major Project: BGSBU Smart Assistant Chatbot – ML-powered university query assistant (84% satisfaction, 5,447 Q&A pairs)

Certifications

- Artificial Intelligence Fundamentals – IBM SkillsBuild
- Data Science & Artificial Intelligence Summer School – IIT Jammu, 2025
- Data Analytics Job Simulation – Deloitte via Forage, Aug 2025
- Gen AI Engineering Mastermind – Outskill, 2025
- Power BI Workshop – OfficeMaster, Feb 2026
- Machine Learning Course – Winnovation, 2024